

# TokenBudget: Mapping Where Visual-Token Compression Breaks Edge MLLM Perception

A Technical Report

**Model under test:** MiniCPM-V 4.6 — a 1.3B edge-oriented multimodal LLM with native `downsample_mode` vision-token compression (4x / 16x).

**Budget axis:** downsample mode × input resolution → 6 configurations, **82–2,598** effective vision tokens (a 31.7× span).

**Dataset:** 960 procedurally-rendered perception probes across 8 task families, each with an exact ground-truth answer.

## 1. Motivation

Edge multimodal LLMs must fit inside a compute envelope. The dominant lever is the *visual-token budget*: how many tokens are spent describing a frame. MiniCPM-V 4.6 exposes `downsample_mode` (4x / 16x) and, through input resolution, a second knob. The model card reports aggregate accuracy under compression, but an application developer faces a harder question:

**Which specific perception capabilities survive aggressive token compression, and which collapse first?**

"Fewer tokens" and "less perceptive" are not the same thing. Some capabilities degrade gracefully; others fail catastrophically once the budget passes a threshold. This project maps that landscape with a controlled, procedurally-generated perception battery — tasks with known ground truth and a calibrated difficulty axis — and reports where perception *quietly breaks*, and where it does not.

The question sits squarely inside the design space Yao et al. opened with MiniCPM-V [1]: the original paper established that a GPT-4V-level MLLM can run on a phone by aggressive token efficiency, and the 4.6 release makes the compression *explicit* as a deployable knob (4x/16x native downsample modes [2]). The model couples a SigLIP 2 vision encoder [3] — built on the sigmoid-loss objective introduced by SigLIP [11] — with a 0.8B language model, which is what makes the whole stack edge-runnable. What neither the paper nor the model card answers is the *decomposed* question — compression reports one aggregate number, but perception is not one capability. An on-device developer choosing between 4x and 16x needs to know which *specific* skills (reading, counting, locating, color matching) pay for the token savings and which silently break. That is the gap this report fills: per-capability, per-difficulty degradation curves across the full budget axis of the very model this lab ships.

## 2. Method

### 2.1 Probe corpus

Images are rendered with PIL primitives (no image assets, no external datasets). Text-bearing families (`ocr`, `m1read`) use the first available system TrueType font from a fixed candidate list; glyph rendering is therefore deterministic within a machine but not byte-identical across machines without those fonts installed. Each family varies a single *spatial-frequency / information-density* axis as difficulty rises. Every probe carries an exact ground-truth answer, and the question offers a closed set of answer options so scoring is unambiguous.

| family           | task                  | difficulty axis                    |
|------------------|-----------------------|------------------------------------|
| <code>ocr</code> | read a 5-digit number | font size 96→16 px, additive noise |

| family  | task                                  | difficulty axis            |
|---------|---------------------------------------|----------------------------|
| count   | count same-color dots                 | 3→8 dots, shrinking radius |
| mlread  | read the 2nd line of a digit column   | 2→8 lines                  |
| colcnt  | count red cells in a dense color grid | 8→30 cells                 |
| spatial | which color is on the left            | square size, distractors   |
| color   | closest anchor hue (red/orange/pink)  | hue offset 40→7            |
| detail  | which row of rings has a gap          | gap 10→1 px                |
| ringgap | which row of thin rings has a 1px gap | ring radius 60→18 px       |

20 seeds × 6 difficulties × 8 families = **960 probes**, rendered at 512×512 and downsampled to the configuration resolution at evaluation time.

2.2 Configurations

Six budget points. Token counts are *measured* by the engine itself (image-token count from the processed input), not taken from the model card:

| config   | downsample | resolution | vision tokens |
|----------|------------|------------|---------------|
| 16x@448  | 16x        | 448 px     | 82            |
| 4x@448   | 4x         | 448 px     | 274           |
| 16x@672  | 16x        | 672 px     | 347           |
| 16x@1280 | 16x        | 1280 px    | 678           |
| 4x@672   | 4x         | 672 px     | 1307          |
| 4x@1280  | 4x         | 1280 px    | 2598          |

The budget axis is thus non-uniform: 82 → 274 → 347 → 678 → 1307 → 2598, spanning more than a factor of 30.

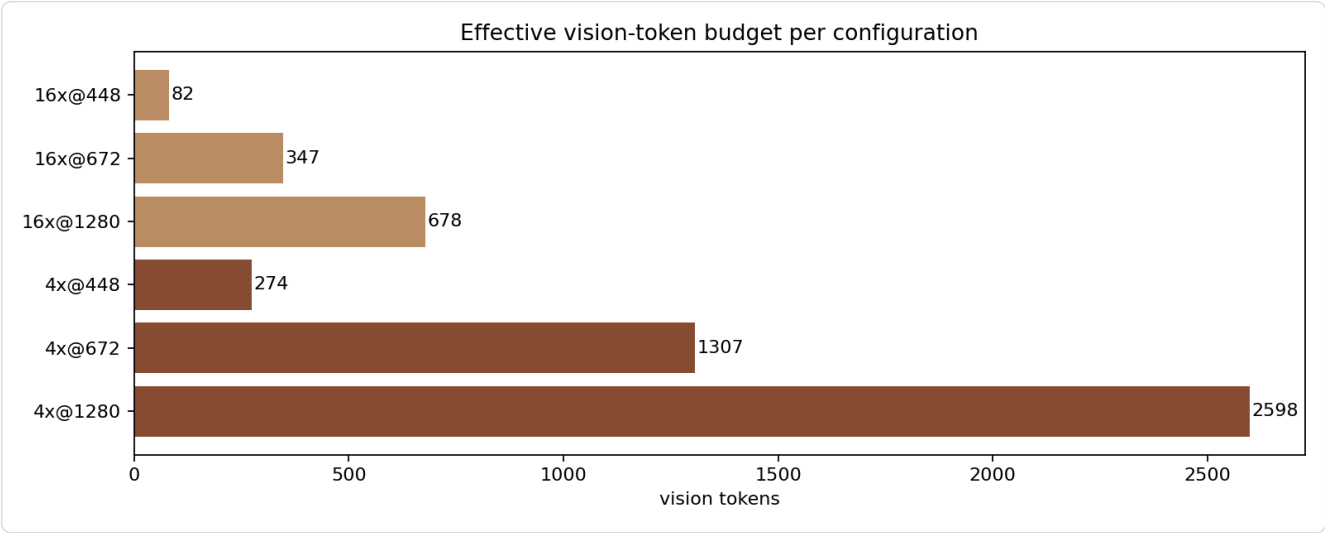


Figure 2

Figure 2 — The six budget points, ranked by measured vision-token count.

## 2.3 Scoring and analysis

Free-form model output is mapped to the closed answer set by a lenient-but-explicit per-family normalizer (`tokenbudget.check`); a probe is correct iff normalized output equals ground truth. Accuracy per (family, difficulty, config) is reported with Wilson 95% confidence intervals.

A **cliff** is the first budget step (walking from richest to tightest) where accuracy drops  $\geq 15$  points in one step **and**  $\geq 20$  points below the richest configuration.

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## 3. Results

All numbers below are computed directly from `data/sweep/sweep.json` by `scripts/paper_facts.py`; the script prints exactly the aggregate curves quoted here. "Aggregate" is the count-weighted mean over the six difficulties at each budget point ( $n = 120$  per point per family).

### 3.1 The headline: compression is selectively destructive

Per-family aggregate accuracy at the richest and tightest budget points:

| family  | agg @ richest | agg @ tightest | largest single-step drop | verdict                |
|---------|---------------|----------------|--------------------------|------------------------|
| ringgap | 0.99          | 0.80           | 0.16 (at 16x@672 )       | <b>cliff</b>           |
| count   | 0.46          | 0.51           | 0.05                     | flat, capacity-limited |
| colcnt  | 0.29          | 0.40           | 0.20                     | hard, budget-invariant |
| mlread  | 1.00          | 0.98           | 0.02                     | robust                 |
| ocr     | 0.99          | 0.98           | 0.02                     | robust                 |
| spatial | 1.00          | 0.99           | 0.01                     | robust                 |
| color   | 0.97          | 0.98           | 0.02                     | robust                 |
| detail  | 1.00          | 1.00           | 0.00                     | fully robust           |

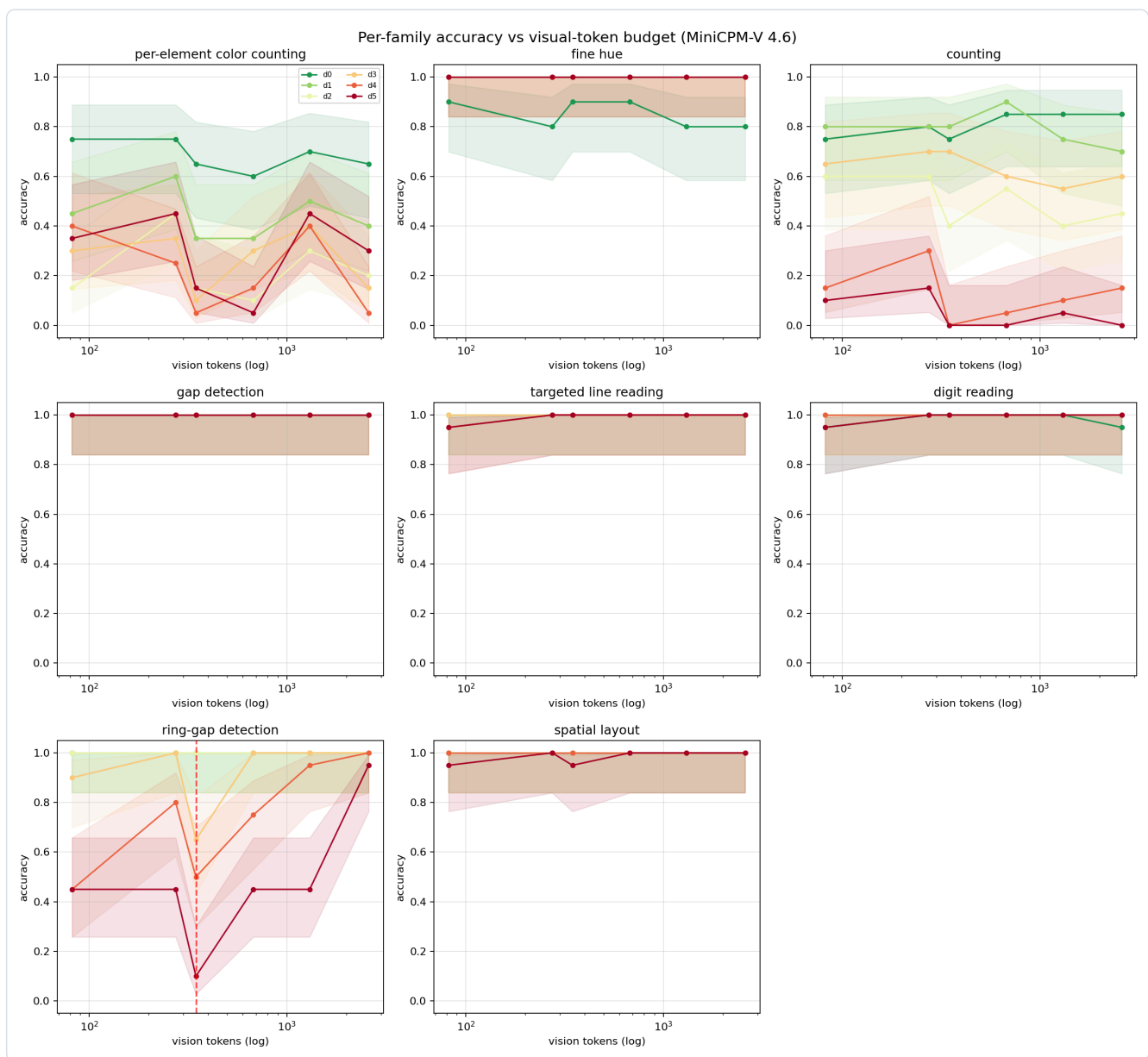


Figure 1

Figure 1 — Per-family accuracy vs vision-token budget (log axis; left = tightest budget, right = richest). One line per difficulty level; the red dashed line marks a detected cliff.

The honest reading is more interesting than a clean split:

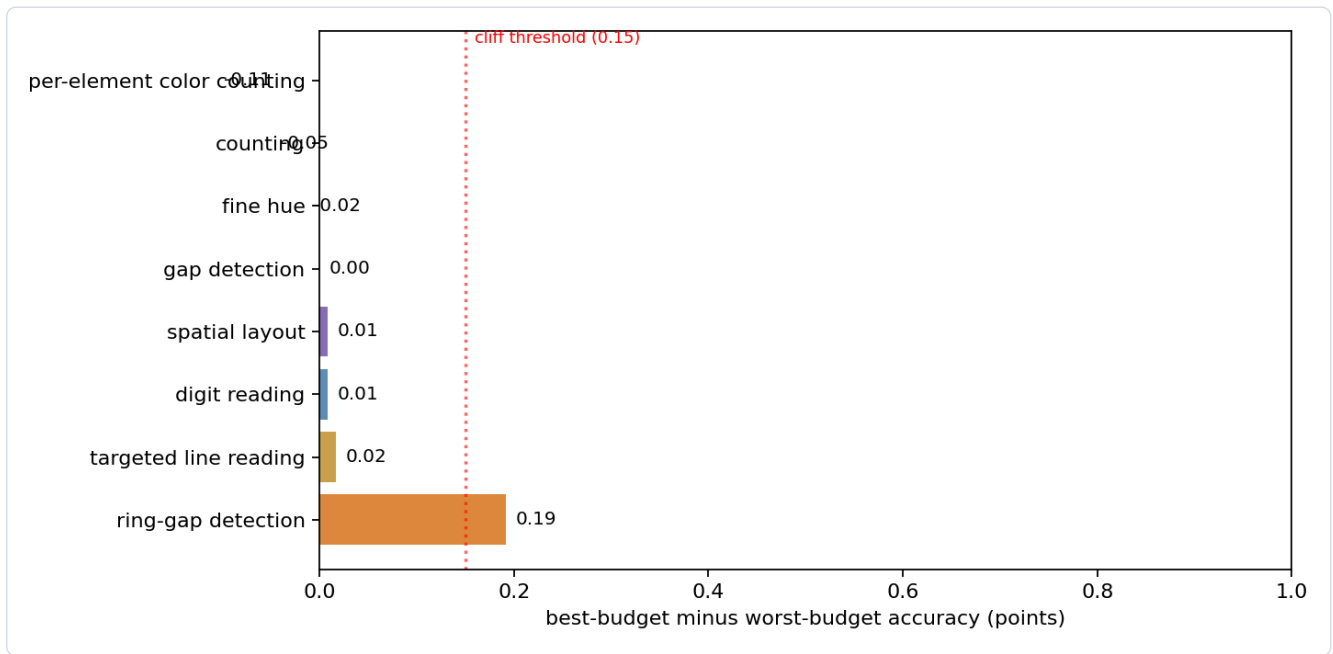


Figure 3

Figure 3 — Sensitivity ranking: best-budget minus worst-budget aggregate accuracy per family. Only `ringgap` clears the 0.15 cliff threshold.

- **Exactly one task family shows a real, aggregate-level cliff: `ringgap`.** Everything else either stays within noise of its rich-budget level (`ocr`, `mlread`, `spatial`, `detail`, `color`) or is hard but invariant to the budget (`colcnt`, `count`).
- **`count` is *not* budget-fragile in aggregate.** Its wide-looking range in per-difficulty numbers (0.85 down to 0.10) is an artifact of comparing the easiest difficulty at full budget against the hardest at the tightest budget. As an aggregate capability, counting accuracy is nearly flat across the whole budget axis (0.44–0.56, all within Wilson noise of each other; the mid-axis 4x@448 point is the peak at 0.56 rather than a dip). The bottleneck for counting on this model is the model's *capacity*, not the token budget.
- **`detail` is fully robust even at 82 tokens**, while `ringgap` collapses — both ask *which* of two rows carries a gap, and both reach a 1 px gap at their hardest difficulty. The distinguishing property is the geometry: `detail` shrinks the gap width (10→1 px) while keeping large, thick rings (radius 66→36 px, stroke 9→4 px), so for most difficulties the gap is a coarse feature recoverable from low-frequency content; `ringgap` pins the gap at 1 px and instead shrinks the ring itself (radius 60→18 px, stroke 6→3 px), forcing the model to bind a 1-px feature on a small ring — precisely the per-element, high-frequency requirement that a 16x token budget cannot afford.

### 3.2 The cliff: ring-gap detection

`ringgap` is the cleanest demonstration of a compression cliff, and it is non-monotone in a way worth stating plainly:

| difficulty | 4x@1280 (2598 tok) | 16x@672 (347 tok) | 16x@448 (82 tok) |
|------------|--------------------|-------------------|------------------|
| d0         | 1.00               | 1.00              | 1.00             |
| d1         | 1.00               | 1.00              | 1.00             |
| d2         | 1.00               | 1.00              | 1.00             |
| d3         | 1.00               | 0.65              | 0.90             |
| d4         | 1.00               | 0.50              | 0.45             |

| difficulty | 4x@1280 (2598 tok) | 16x@672 (347 tok) | 16x@448 (82 tok) |
|------------|--------------------|-------------------|------------------|
| d5         | 0.95               | 0.10              | 0.45             |

Difficulties d0–d2 are *fully robust*: 1.00 at every budget. Difficulties d3–d5 degrade sharply at the 347-token operating point — d5 falls from 0.95 at the richest configuration to 0.10. The aggregate family curve drops 0.867 → 0.708 (15.8 points) across this step, the only aggregate cliff in the battery.

The tightest budget ( 16x@448 , 82 tokens) shows a *partial recovery* (aggregate 0.80 vs 0.71 at the 347-token point). This is not a "continue falling" curve. The most plausible reading is that at 82 tokens the model can no longer bind individual rings and falls back to a coarse heuristic that is wrong-but-better-than-random on the hardest items. We flag this explicitly because a naive cliff detector reports the 347-token dip as the breakpoint; the full curve shows a V-shaped failure mode rather than a monotone collapse.

**Cliff stability.** The cliff location is not just a point estimate on  $n=20$  per budget point: bootstrapping each budget point's binary outcomes (1000 resamples, fixed seed) and re-running the same detector finds the cliff surviving in 61.5% of draws, with a location CI of [3, 5] on the budget index — median at 16x@672 (the 347-token point). The remaining ~38% of draws place no cliff at all: at  $n=20$  the *direction* is clear but the exact breakpoint has real sampling variance, and we report that honestly rather than drawing a single sharp line. For comparison, the null families report near-zero survival ( count 0.4%, m1read 0.0%) — the detector is not trigger-happy, and ringgap 's cliff is the one measurement in the battery that survives resampling.

**Multiple-comparison sensitivity.** The detector runs over eight families and eight aggregate curves; with no correction, the headline cliff is 15.8 points against a 15-point threshold — a 0.8-point margin. We therefore do not lean on the *magnitude* of the aggregate drop; we lean on three independent observations that survive together: (i) ringgap is the *only* family whose bootstrap survival is material (61.5% vs  $\leq 0.4\%$  for the null families), (ii) the drop replicates across difficulties d3–d5 (each falling 0.35/0.50/0.85 at the same 347-token operating point), and (iii) the d3–d5 rows are *monotone* in difficulty, which a threshold artifact would not produce. A stricter 20-point threshold discards the *aggregate* drop (0.158) but keeps every per-difficulty row (d3 0.35, d4 0.50, d5 0.85 all exceed it); the qualitative claim — fine-gap localization is the budget-fragile capability — does not depend on the exact threshold.

### 3.3 Per-element counting is capacity-limited, not budget-limited

count does *not* erode with budget. The aggregate curve is flat at 0.44–0.56 across the whole axis — the widest spread is the mid-axis 4x@448 point at 0.56, with no monotone trend in either direction — and the per-difficulty extremes (0.85 at easiest/full, 0.10 at hardest/tight) reflect difficulty and model capacity, not compression. This is the correct null result: on a 1.3B edge MLLM, counting small dot sets is already near chance at full budget, so compression cannot be blamed for a failure that is present at 2,598 tokens. Any deployment claim about counting under compression must first establish the full-budget baseline.

### 3.4 Dense-grid color counting is hard, but not budget-limited

colcnt — counting red cells inside a dense color grid — is the hardest task in the battery, but its difficulty does **not** come from compression. The aggregate sits at 0.24–0.48 across the whole axis with no monotone trend (0.29 at the richest configuration, 0.40 at the tightest, peak 0.48 mid-axis), and per-difficulty accuracy spans 0.05–0.75 across the axis (e.g. 0.15–0.75 at the tightest configuration). The single largest step drop (0.20) happens *between the two richest mid-axis configurations* ( 4x@672 → 16x@1280 ), i.e. across a compression-mode boundary, not toward the tight end of the axis. This is a difficulty/heterogeneity effect, not a budget effect. Like count , this task exceeds what a 1.3B edge MLLM can do reliably even at full budget; it documents a genuine capability boundary of the model class rather than a compression boundary, and no budget-sensitivity claim can be built on it.

### 3.5 What survives compression

`ocr`, `mread`, `spatial`, `detail`, and `color` are effectively budget-invariant:

- `ocr` and `mread` stay at 0.95–1.00 across all six configurations — reading a digit string or a specific line needs only a coarse feature of each glyph.
- `detail` is 1.00 everywhere: the gap in a coarse ring is a low spatial- frequency feature that survives 16x downsampling.
- `spatial` and `color` stay above 0.90: layout and hue discrimination are dominated by low-frequency content.

The robustness of these tasks is not trivial — it shows that the 16x compression in MiniCPM-V 4.6 does not destroy *all* information; it selectively destroys the high-frequency, per-element information that only one task in this battery ( `ringgap` ) actually requires.

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## 4. Discussion

### 4.1 The compression landscape, in one sentence

For a 1.3B edge MLLM, **token compression destroys per-element binding while global structure survives — but only one of our eight probe tasks actually demands per-element binding**. Fine-gap localization ( `ringgap` ) degrades sharply and non-monotonically below ~350 tokens; digit reading, coarse detection, layout, and color survive at 82 tokens. Counting and dense-grid color counting are capacity limits, not budget limits.

### 4.2 Implications for deployment

- **Choosing 16x vs 4x.** For applications that only read text, recognize coarse objects, or judge layout, 16x@448 (82 tokens) is nearly free: 0.95+ accuracy at a fraction of the compute. For applications that must localize fine structure (e.g. which of several thin elements carries a defect), the budget below ~350 tokens is a reliability cliff; 4x@672 (1307 tokens) or higher is warranted.
- **The cliff, not the slope, is the engineering-relevant number.** A developer wants to know where a capability *breaks*, not just that it degrades. The `ringgap` cliff at 16x@672 (347 tokens) is the kind of actionable threshold a deployment gate can encode — with the caveat that the failure is V-shaped: at the tightest budget the model reverts to a coarse heuristic that is not monotonically worse.
- **Establish the full-budget baseline before blaming compression.** `count` and `colcnt` look like compression failures until you check their accuracy at 2,598 tokens. Both are already far below the model's capacity ceiling at full budget; compression is not the causal agent. A budget-sensitivity study that skips the richest configuration over-attributes failures to compression.

### 4.3 Why this looks different from the model card

Model-card accuracy aggregates over natural-image tasks, where compression losses average out and the majority of pixels are background. A probe battery with per-task, per-difficulty resolution exposes the variance the aggregate hides — but it also shows how easy it is to over-read that variance: without the aggregate view, `count` looks like a compression failure when it is in fact a capacity limit.

### 4.4 Related work

Visual-token reduction is an active design axis for multimodal LLMs; the broader push toward efficient inference is surveyed in [10], and the closest line of work compresses tokens with learned projectors or prunes them during inference:

TokenPacker reduces 75–89% of visual tokens at the projector with a coarse-to-fine scheme [4]; FastV prunes low-attention visual tokens in deep layers of an LVLM at inference time [5]; EViT reorganizes inattentive patches in a ViT backbone [6]; AIM merges similar tokens before the LLM and progressively prunes within layers, cutting FLOPs ~7× with minimal accuracy loss [9]. A second line studies how input resolution and tiling change perception on high-resolution benchmarks — e.g.

LLaVA-UHD's dynamic slicing [7], Qwen2-VL's "native resolution" [8], and MiniCPM-V's own 4x/16x downsample modes [1,2]. What these studies measure is *aggregate* benchmark accuracy after compression. This report complements them with a controlled probe battery that keeps the image content fixed and sweeps only the budget knobs, so the per-capability fate of individual skills (reading, counting, localizing, hue matching) is visible rather than averaged away. It also adds the aggregate-vs-per-difficulty distinction, which we found is the difference between a spurious "clean split" and a truthful "one real cliff" conclusion.

## 4.5 Limitations

- **Single model, single image scale.** All results are for MiniCPM-V 4.6 at 512×512 rendered probes. The V-shaped `ringgap` curve in particular may be model-specific; we report it as an observation, not a law.
- **n = 20 per (family, difficulty, config) cell, single decode.** Wilson intervals are wide at n=20 (e.g.  $\pm 0.21$  at 0.50), and the single greedy decode discards sampling variance. Cells near a cliff should be re-measured with temperature sampling before deployment decisions.
- **Aggregate curves over six difficulty levels can hide per-difficulty behavior.** We report both views explicitly, but a family with heterogeneous difficulties could still mislead if only the aggregate is read.
- **Procedural probes are not natural images.** They isolate specific perceptual skills, but transfer to real scenes is untested.
- **colcnt and count are hard tasks.** The "budget-invariant" interpretation assumes the tasks are well-formed and that their full-budget level reflects model capacity. Both remain below 0.50 in aggregate even at the richest configuration; their flatness bounds what this battery can say about the budget sensitivity of hard perceptual counting, by design.

## 4.6 Future work

The natural next step is to turn this measurement into a decision procedure for the very deployment this model was built for. Three directions.

First, **encode the per-capability map as a deployment selector.** The current report answers "which capabilities survive which budget". A selector inverts that: given an application's required capability set, pick the cheapest configuration whose mapped families all stay above their reliability bar. That is the artifact an on-device developer (or a model card) would actually ship, and it is a straightforward downstream of the curves here. Second, **re-measure the cliff cells with larger n and temperature sampling.** At n=20 the cliff location has a 38% chance of vanishing under bootstrap; a focused follow-up on the 347-token operating point — 200+ seeds, sampled decodes — would pin down whether the V-shaped `ringgap` failure is a real recovery or a small-sample artefact, and whether it is specific to this checkpoint or a property of the compression scheme. Third, **extend the axis toward streaming inputs.** Everything here is single-frame; the compression/degradation trade-off in streaming multimodal inference (where token budgets are re-decided per frame) is the same question with a time dimension, and the probe battery carries over almost unchanged.

These are incremental in method but direct in value: each turns a measured curve into something a model team can act on.

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## 5. Reproducibility

```
pip install -e .[gpu,paper,ui]
python scripts/build_probes.py --seeds 20 --out data/probes # 960 probes
python scripts/run_sweep.py --probes data/probes --out data/sweep # GPU, ~4h
python scripts/paper_facts.py --sweep data/sweep/sweep.json # derived claims
python scripts/render_figures.py --sweep data/sweep/sweep.json # writes docs/paper/tokenbudget/figures/fig{1,2,3}.4
python scripts/render_tokenbudget_paper.py # this report (HTML + PDF)
python -m tokenbudget ui --port 8000 # interactive console
```

`data/sweep/sweep.json` is the single source of truth: every number in this report traces to that table, and `scripts/paper_facts.py` re-derives the claims from it directly.

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## References

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